

Day 11 Hybrid Notes

Right triangles, distance formula, and radicals | Cloze + outline + active learning

Name:	Date:	Period:
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Learning Targets
I can use the Pythagorean theorem to find missing side lengths.
I can use the converse to decide whether a triangle is right.
I can simplify common square roots in geometry problems.

Part 1: Cloze Notes - Essential Ideas
1. The Pythagorean theorem is $a^2 + b^2 =$ _____.
2. The hypotenuse is the longest side of a right _____.
3. The hypotenuse is across from the right _____.
4. The converse checks whether a triangle is _____.
5. The distance formula comes from the Pythagorean _____.
6. $\sqrt{50}$ simplifies to _____.
7. Always square side lengths before _____.

Part 2: Outline Notes - Big Picture A. Right triangles connect geometry to _____. B. Distance on a coordinate plane creates a right _____. C. Radical simplification keeps exact answers _____. D. The converse helps classify _____.	Connection to Prior Math 1. What algebra skill shows up today? _____ 2. What diagram or graph skill shows up today? _____ 3. What is one place this could appear outside math class? _____ —
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Most Important Reminder
Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

Day 11 Hybrid Notes

Guided setup + active learning

Part 3: Guided Setup Practice

1. Find the hypotenuse when the legs are 6 and 8.

Setup/work: _____

Answer: _____

2. Simplify $\sqrt{72}$.

Setup/work: _____

Answer: _____

Part 4: Error Analysis

A student solves $5^2 + c^2 = 13^2$ when 13 is a leg.
What went wrong?

Correct idea:

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Before You Solve Checklist

☐ I labeled the diagram or graph.

☐ I chose the correct relationship or formula.

☐ I substituted values carefully.

☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that

_____.

Sentence 2: One thing I need to remember when solving is

_____.

Part 5C: Application Problem

A park path goes 9 blocks east and 12 blocks north. What is the straight-line distance?

Setup: _____

Answer: _____

Sentence answer: _____